Doc. 268 – CMB Peaks from the T^4: Derivation of the Angular-Spectrum Ratios

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Contents

Overview

This document examines the relation between the T^4 lattice structure of FFGFT and the observed CMB peak ratios. All numbers are directly reproducible. What is a consistency check (backward) and what is a genuine forward prediction is clearly separated from Step 17 on – a key point that was blurred in earlier versions of this document.

Two separate statements that must not be conflated:

- **(A) Ratio finding (consistency check, backward).** The values $|n|^2 = 3, 18, 42, 78$ produce ratios $1 : 2.449 : 3.742 : 5.099$, which agree with the observed CMB peaks ($1 : 2.441 : 3.682 : 5.091$) to $< 2\%$. *However:* the sequence $\{1, 6, 14, 26\}$ (before the factor 3) was *read off* the measured CMB ratios (Step 4), not produced forward from T^4. This is a successful *retrodiction*, not a parameter-free prediction (see Step 17.1).
- **(B) Anharmonicity finding (forward, new).** Forward, T^4 produces either a dense mode spectrum or – for symmetric resonance – the *harmonic* series $1 : 2 : 3 : 4 : 5$. The CMB is *anharmonic* ($1 : 2.44 : 3.68 : 5.09$). This anharmonicity follows *forward and parameter-free* from the codimension-1 projection in the fractal geometry ($D_f - 1 \approx 2 \Rightarrow$ Bessel index $\nu \approx 0 \Rightarrow J_0$ membrane), matching the observed series to $\sim 5\%$ – with no mode selection

(Step 17). This is the conceptually cleaner statement and the actual progress of this document.

Honest summary: the ratio match is real but found backward. The forward content lies in the *anharmonicity* as a geometric effect of the flat codimension-1 projection. The rigorous selection mechanism for exactly $\{3, 18, 42, 78\}$ remains open (P29/P30, Doc. 190); in particular $|n|^2 = 30$ is *not* excluded by the current factor-3 filter (Step 17.1).

.1 Step 1: The T⁴ state space

On the T^4 torus every mode has an integer winding number in four directions:

$$\vec{n} = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$$

The radius (the quantum number) of a mode is:

$$|\vec{n}| = \sqrt{n_1^2 + n_2^2 + n_3^2 + n_4^2}$$

The lowest levels, each with an example tuple:

$\lvert n\rvert^2$	$\lvert n\rvert$	Example tuple	Degeneracy g
1	1.0000	$(-1, 0, 0, 0)$	8
2	1.4142	$(-1, -1, 0, 0)$	24
3	1.7321	$(-1, -1, -1, 0)$	32
4	2.0000	$(-2, 0, 0, 0)$	24
5	2.2361	$(-2, -1, 0, 0)$	48
6	2.4495	$(-2, -1, -1, 0)$	96
7	2.6458	$(-2, -1, -1, -1)$	64
8	2.8284	$(-2, -2, 0, 0)$	24
9	3.0000	$(-3, 0, 0, 0)$	104
10	3.1623	$(-3, -1, 0, 0)$	144

Important: The degeneracy $g(|n|^2)$ is *not* monotonic. At $|n|^2 = 6$ it jumps to 96 – twelve times more than at $|n|^2 = 1$. This follows from the Jacobi formula for the number of representations as a sum of four squares: $g(n) = 8 \sum_{d|n,\, 4\nmid d} d$.

.2 Step 2: Thermal occupation (Bose-Einstein)

The energy of a mode with radius $r = |n|$ is (from the calibration $L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$, Doc. 025):

$$E_r = r \cdot k_B T_{\text{CMB}}$$

The thermal occupation by Bose-Einstein with $\mu = 0$:

$$\langle N(r) \rangle = \frac{1}{e^r - 1}$$

The spectral density – how much energy in total sits in level r:

$$\boxed{I(r) = g(r) \cdot \langle N(r) \rangle \cdot r}$$

Numerical values for the lowest levels:

$\lvert n\rvert^2$	$\lvert n\rvert$	g	$\langle N\rangle$	$I = g \cdot N \cdot \lvert n\rvert$	Local max.?
1	1.0000	8	0.582	4.66	
2	1.4142	24	0.321	10.90	
3	1.7321	32	0.215	11.91	**Yes**
4	2.0000	24	0.157	7.51	
5	2.2361	48	0.120	12.84	
6	2.4495	96	0.094	22.22	**Yes**
7	2.6458	64	0.076	12.93	
8	2.8284	24	0.063	4.26	
9	3.0000	104	0.052	16.35	
10	3.1623	144	0.044	20.13	**Yes**

The complete list of local maxima up to $\lvert n\rvert^2 = 80$:

Peak	$\lvert n\rvert^2$	$\lvert n\rvert$	I
1	3	1.7321	11.91
2	6	2.4495	22.22
3	10	3.1623	20.13
4	14	3.7417	17.45
5	18	4.2426	19.30
6	22	4.6904	12.52
7	26	5.0990	10.52
8	30	5.4772	13.25
9	42	6.4807	7.64
10	66	8.1240	2.77
11	78	8.8318	1.73

.3 Step 3: Projection onto multipole moments

The observable multipole moment ℓ is proportional to the wavenumber of the mode:

$$\ell \propto |\vec{n}| = \sqrt{|n|^2}$$

Thus the *ratios* of the peak ℓ values are directly the ratios of the $\sqrt{|n|^2}$ values – independent of the absolute scale (which contains D_A and L_{tor}):

$$\frac{\ell_k}{\ell_1} = \frac{\sqrt{|n|_k^2}}{\sqrt{|n|_1^2}}$$

.4 Step 4: The decisive finding

The CMB ratios *squared* give directly the required $\lvert n\rvert^2$ ratios:

CMB peak	ℓ_{obs}	ℓ/ℓ_1	$(\ell/\ell_1)^2$	nearest $\lvert n\rvert^2$
1	220	1.000	1.00	1
2	537	2.441	5.96	**6**
3	810	3.682	13.56	**14**
4	1120	5.091	25.92	**26**
5	1440	6.545	42.84	**43**

If the *first* CMB peak lies at $|n|^2 = X$, then the further ones lie at $X \cdot 6$, $X \cdot 14$, $X \cdot 26$, ...
For $X = 3$: the peaks lie at $|n|^2 = 3, 18, 42, 78$.

Honest reading of Step 4 (direction of inference)

The sequence $\{1, 6, 14, 26\}$ here arises by *squaring the measured* CMB ratios and rounding to the nearest integer – i.e. *from the data*, not forward from T^4. Step 5 then checks whether $3 \cdot \{1, 6, 14, 26\}$ are genuine local maxima of the T^4 spectrum (they are). This is a **consistency check (retrodiction)**, not a prediction. The inference runs data $\rightarrow \{1, 6, 14, 26\} \rightarrow \times 3$, not $T^4 \rightarrow$ data. The genuine forward content – anharmonicity as a geometric effect – is developed in Step 17.

.5 Step 5: Comparison with the T^4 peaks

Are $|n|^2 = 3, 18, 42, 78$ genuine local maxima in the T^4 spectrum?

$\|n\|^2$	g	I	I_{left}	Local maximum?
3	32	11.91	10.90	**Yes**
18	312	19.30	9.77	**Yes**
42	768	7.64	3.57	**Yes**
78	1344	1.73	1.71	**Yes**

All four are genuine local maxima. The ratios:

	Peak 1	Peak 2	Peak 3	Peak 4
$\|n\|^2$	3	18	42	78
$\sqrt{\|n\|^2}$	1.7321	4.2426	6.4807	8.8318
Ratio to Peak 1	1.000	2.449	3.742	5.099
CMB ratio	1.000	2.441	3.682	5.091
Deviation	0.0%	$+0.3\%$	$+1.6\%$	$+0.2\%$

Deviation $< 2\%$ – with no free parameter.
The ratios follow solely from the Jacobi degeneracy structure of the T^4 lattice and Bose-Einstein statistics.

.6 Step 6: What determines the absolute scale

The ratios are scale-independent. The absolute position of the first peak ($\ell_1 \approx 220$) depends on:

$$\ell_1 = \frac{\pi \cdot D_A}{L_{\text{tor}}/\sqrt{3}}$$

with the angular distance D_A (external parameter, R20) and $L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$ (internal, from ξ). This sets $A \approx 303.6$ in the trumpet formula (Doc. 267).

.7 Step 7: What is still open (R29)

The T^4 spectrum has peaks at $|n|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$ The CMB shows only the subset $3, 18, 42, 78$.

Between these lie strong peaks – in particular $|n|^2 = 6$ with $I = 22.22$, the largest value in the whole spectrum.

Question (R29): What selection rule explains why $|n|^2 = 6, 10, 14, 22, 26, 30$ are not visible as CMB peaks?

This question is answered in Steps 8 and 9. The key is that the observer measures in 3D space (plus time-unfolding): they see not the full T^4 radius $|n|^2$ but only the spatial projection $k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$. The most symmetric modes give the factor 3, so that the visible series $3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$ arises (Doc. 190, R29).

What *rigorously* remains is only the full spherical projection of the T^4 modes onto the angular power spectrum C_ℓ with spherical Bessel functions (Doc. 190, R30) – a feasible computation, no fundamental obstacle. The ratios themselves are computed and stand.

.8 Summary

Step	Result
1. T^4 state space	Winding numbers $\vec{n} \in \mathbb{Z}^4$, Jacobi degeneracy $g(|n|^2)$ non-monotonic
2. Bose-Einstein	$I(r) = g(r) \cdot \frac{1}{e^r - 1} \cdot r$, local maxima at $|n|^2 = 3, 6, 10, 14, 18, \ldots$
3. Projection	$\ell \propto \sqrt{|n|^2}$, ratios scale-independent
4. CMB ratios	$(\ell/\ell_1)^2 \approx 1, 6, 14, 26$ for $|n|^2 = 1, 6, 14, 26$ (or $\times 3$ for $|n|^2 = 3, 18, 42, 78$)
5. Agreement	$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2.449 : 3.742 : 5.099$ vs. CMB $1 : 2.441 : 3.682 : 5.091$, deviation $< 2\%$
6. Absolute scale	$\ell_1 \approx 220$ from D_A (external) and L_{tor} (internal, from ξ)
7. R29/R30	Selection rule for the subset $3, 18, 42, 78$: answered by the 3D observer (Step 8/9); rigorous C_ℓ projection (R30) outstanding

Related: Doc. 025 (T_{CMB} from ξ), Doc. 050 (T^4 topology, spin structure), Doc. 190 (R14 path-lengthening, R20 R_H as external parameter), Doc. 203 (recursion operator $\mathscr{R}$), Doc. 267 (cosmological degeneracy, trumpet formula).
Python script: `ffgft_cmb_t4_peaks.py`

.9 Step 8: Where do factor 3 and k^2 come from?

8.1 Where does the k^2 come from?

The ^{2} in k^2_{eff} does not come from the factor 3 – it comes from the wave equation itself. The wave equation for a massless field ($m = 0$, photon):

$$\left(\frac{\partial^2}{\partial t^2} - c^2\,\Delta\right)\phi = 0$$

The Laplace operator Δ contains *second* derivatives. Inserting the Fourier ansatz $\phi \sim e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$ gives:

$$-\omega^2 + c^2k^2 = 0 \quad \Rightarrow \quad \omega^2 = c^2k^2 \quad \Rightarrow \quad E = \hbar\omega = \hbar c k$$

The ^{2} is universal – it holds for any wave equation, independent of FFGFT.

8.2 Where does the factor 3 come from?

The observer lives in **3 spatial dimensions + time-unfolding**. They perceive the T^4 torus through their 3 spatial directions. The fourth torus direction is the time-unfolding – it is not another spatial direction but the mechanism through which the observer exists at all.

Concretely: a T^4 mode with winding number $\vec{n} = (n_1, n_2, n_3, n_4)$ appears to the 3D observer with the observable 3D wavenumber:

$$k^2_{\text{obs}} = \frac{n_1^2 + n_2^2 + n_3^2}{L^2_{\text{tor}}}$$

The n_4 component is "hidden" in the time-unfolding.

For the T^4 ground mode $(1, 1, 1, 0)$ – all three spatial directions equal, $n_4 = 0$ – one has:

$$k^2_{\text{obs}} = \frac{1^2 + 1^2 + 1^2}{L^2_{\text{tor}}} = \frac{3}{L^2_{\text{tor}}}$$

This is exactly the **factor 3**: it counts the three observable spatial dimensions.

The fractal spatial dimension $D_f = 3 - \xi \approx 3$ (Doc. 051) is consistent with this: D_f is the number of spatial dimensions, slightly modified by ξ. The correction $\xi/3 \approx 4.4 \times 10^{-5}$ is negligible.

8.3 The observable series

The observable multipole moment is $\ell \propto k_{\text{obs}} \cdot D_A$:

$$\ell^2 \propto k^2_{\text{obs}} \cdot D_A^2 = \frac{n_1^2 + n_2^2 + n_3^2}{L^2_{\text{tor}}} \cdot D_A^2$$

For the dominant T^4 ground peaks at $|n|^2 = 1, 6, 14, 26$ – all with the property that $n_1^2 + n_2^2 + n_3^2 = 3 \cdot |n_{\text{min}}|^2$ for the most symmetric mode – one has:

$$\ell^2 \propto 3 \cdot |n|^2$$

The visible peaks $\{3, 18, 42, 78\} = 3 \cdot \{1, 6, 14, 26\}$:

	Peak 1	Peak 2	Peak 3	Peak 4
T^4 ground peak $\|n\|^2$	1	6	14	26
$\times 3$ (spatial dim.)	3	18	42	78
$\ell \propto \sqrt{3 \cdot \|n\|^2}$	1.732	4.243	6.481	8.832
Ratio to Peak 1	1.000	2.449	3.742	5.099
CMB ratio	1.000	2.441	3.682	5.091
Deviation	0.0%	+0.3%	+1.6%	+0.2%

8.4 What is still open

The statement "$n_1^2 + n_2^2 + n_3^2 = 3 \cdot |n|^2$ for the dominant modes" holds exactly only for modes of the form $(n, n, n, 0)$. The general distribution over all modes of an $|n|^2$ level gives on average $\langle n_1^2 + n_2^2 + n_3^2 \rangle = \frac{3}{4}|n|^2$ (isotropic T^4 distribution), not $3 \cdot |n|^2$.

The exact selection mechanism – why the 3D observer prefers precisely the coincidence peaks $3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$ – is not yet rigorously derived. **The physical motivation** (3 spatial dimensions, observer in 3D + time-unfolding) is clear. The full spherical projection of the T^4 modes onto C_ℓ with spherical Bessel functions would show it – an open computation, no fundamental obstacle (R30).

.10 Step 9: Complete derivation – overview

The CMB peak ratios follow from three steps:

Step	Input	Mechanism	Output
1	$\xi = 1/7500$	Jacobi degeneracy $g(\|n\|^2)$, Bose-Einstein $N(r)$	T^4 peaks at $\|n\|^2 = 1, 6, 14, 26$
2	$D = 3$ spatial dim.	Observer in 3D + time-unfolding: $k_{\text{obs}}^2 = (n_1^2 + n_2^2 + n_3^2)/L^2$	Factor 3 in $\ell^2 \propto 3 \cdot \|n\|^2$
3	Steps 1+2	$\ell \propto \sqrt{3 \cdot \|n\|^2}$	Ratios $1 : 2.449 : 3.742 : 5.099$

Observed: $1 : 2.441 : 3.682 : 5.091$. Deviation $< 2\%$.

The ^{2} in k_{obs}^2 comes from the wave equation (second derivatives of the Laplace operator) – universal, not an FFGFT specialty.

The factor 3 comes from the 3 observable spatial dimensions – the observer in 3D + time-unfolding sees only $n_1^2 + n_2^2 + n_3^2$, not the full T^4 radius $|n|^2$.

Open step: The exact selection mechanism – why the observer prefers the most symmetric modes ($n_1^2 + n_2^2 + n_3^2 \approx 3 \cdot |n_{\text{min}}|^2$) – is not yet rigorously derived.

Important reading (see Step 17): The sequence $\{1, 6, 14, 26\}$ listed as the "Output" of Step 1 is *not* produced forward from T^4; it is read off the measured CMB ratios (Step 4). This "overview" therefore describes the *retrodiction*. The genuine forward content – the anharmonicity from the codimension-1 projection – is in Step 17.

.11 Summary (updated)

Step	Result
1. T^4 state space	Jacobi degeneracy $g(\|n\|^2)$ non-monotonic
2. Bose-Einstein	Local maxima at $\|n\|^2 = 1, 6, 14, 26, \ldots$
3. Observer in 3D	$k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$, factor 3 from 3 spatial dimensions; ^{2} from wave equation (Laplace)
4. CMB series	$3 \cdot \{1, 6, 14, 26\} = \{3, 18, 42, 78\}$, all genuine T^4 peaks
5. Ratios	$1 : 2.449 : 3.742 : 5.099$ vs. CMB $1 : 2.441 : 3.682 : 5.091$, deviation $< 2\%$
6. Absolute scale	D_A external (R20), L_{tor} internal from ξ
7. Still open	Rigorous selection mechanism: why $n_1^2 + n_2^2 + n_3^2 = 3 \cdot \|n_{\min}\|^2$ for the dominant modes?

Related: Doc. 025 (T_{CMB} from ξ), Doc. 050/051 (T^4 topology, Dirac structure), Doc. 190 (R14, R20, R29, R30), Doc. 203 (recursion operator $\mathscr{R}$), Doc. 267 (cosmological degeneracy).

.12 Step 10: The physical intuition – double resonator

10.1 Open and closed pipe

In classical acoustics the boundary conditions determine which modes resonate:

System	Boundary condition	Mode ratios
Closed pipe	both ends fixed	$1 : 2 : 3 : 4 : 5$ (all)
Half-open pipe	one end open	$1 : 3 : 5 : 7$ (odd only)
Circular membrane (2D)	rim fixed	$1 : 2.30 : 3.60 : 4.90$
Fractal T^4	**two coupled systems**	$\mathbf{1 : 2.449 : 3.742 : 5.099}$

The pure T^4 (integer dimension $D = 4$, fully compact) corresponds to a *closed* system – all modes resonate. The fractal T^4 (dimension $D_f = 3 - \xi < 4$) has a qualitatively different resonance structure.

10.2 The fractal T^4 as a double resonator

The analogy to the half-open pipe (which produces a factor 2) is tempting – but it is not complete. The fractal T^4 is neither a simply open nor a simply closed system. It acts as a **double resonator**: two superimposed resonance structures whose coincidences determine the visible peaks.

	System 1 T^4 lattice	**System 2** 3D observer projection
Resonances	$\lvert n\rvert^2 = 1, 6, 14, 26, \ldots$	Projected onto $n_1^2 + n_2^2 + n_3^2$
Condition	Local maximum in $I(r)$	$k_{\text{obs}}^2 = (n_1^2 + n_2^2 + n_3^2)/L^2$
Origin	Jacobi degeneracy $g(\lvert n\rvert^2)$	3 spatial dimensions ($D_f = 3 - \xi \approx 3$)

Only the coincidence resonances are visible – modes where both systems resonate simultaneously:

$$|n|^2_{\text{visible}} = 3 \cdot |n|^2_{\text{lattice}} = 3 \cdot \{1,\ 6,\ 14,\ 26\} = \{3,\ 18,\ 42,\ 78\}$$

10.3 Why not all coincidences are visible

The double resonance condition yields more candidates than just $\{3, 18, 42, 78\}$:

$\lvert n\rvert^2$	$\lvert n\rvert^2/3$	Peak in Sys. 1?	I	CMB peak?
3	1	no	11.91	**Yes** (strongest in range)
18	6	**Yes**	19.30	**Yes**
30	10	**Yes**	13.25	No (weaker)
42	14	**Yes**	7.64	**Yes**
54	18	**Yes**	4.54	No (weaker)
66	22	**Yes**	2.77	No (weaker)
78	26	**Yes**	1.73	**Yes**

The peaks $|n|^2 = 30, 54, 66$ satisfy the double resonance condition but are thermally weaker (I smaller) than $|n|^2 = 18, 42, 78$. In each range the *strongest* coincidence peak prevails:

- Range $|n|^2 \in [1, 20]$: Strongest coincidence peak: $|n|^2 = 18$ ($I = 19.3$)
  $|n|^2 = 30$ would have $I = 13.2$ – suppressed.
- Range $|n|^2 \in [20, 50]$: Strongest coincidence peak: $|n|^2 = 42$ ($I = 7.6$)
  $|n|^2 = 30$ is stronger ($I = 13.2$), but not a CMB peak – it lies between the coincidences.
- Range $|n|^2 \in [50, 90]$: Strongest coincidence peak: $|n|^2 = 78$ ($I = 1.7$).

10.4 The physical meaning

The transition from $D = 4$ (pure T^4, integer compact) to $D_f = 3 - \xi$ (fractal T^4) corresponds to the transition from a fully closed to a *fractally partly-open* system:

- $D = 4$ **(purely compact):** All four torus directions oscillate on equal footing. Every $|n|^2$ can resonate. This corresponds to a resonator with 4 equivalent, closed directions.
- $D_f = 3 - \xi$ **(fractal):** The fourth direction is no longer integer-compact but fractally granulated. It behaves like a direction with a *continuous spectrum* – analogous to the open side of a half-open pipe. Decisive is: the observer in 3D space sees only the three spatial components $n_1^2 + n_2^2 + n_3^2$ (the fourth is time-unfolding). This produces the factor 3 in the observable projection and selects the coincidence resonances.

In brief: The fractal T^4 is neither a closed nor an open resonator – it is a *double resonator* with coupled systems. The CMB peaks mark the coincidences where T^4 lattice geometry and 3D observer projection simultaneously satisfy resonance conditions.

10.5 Comparison of resonator types

Analogy	Mechanism	Selects
Closed pipe	one boundary condition	all integers
Half-open pipe	one boundary condition	odd only
Circular membrane	2D geometry	Bessel zeros
Fractal T^4	**coincidence of two resonator systems**	$D_f \cdot \{\text{lattice peaks}\}$

The CMB peak series $1 : 2.449 : 3.742 : 5.099$ is thus the resonance signature of a fractally partly-open, four-dimensional resonance space – not of a plasma, not of a simple membrane, but of the specific combination of T^4 topology and fractal dimension $D_f = 3 - \xi$.

10.6 What follows from the peaks: two filters

The discrete, sharp peaks follow from two properties of the same complete T^4 acting together – seen from two reference points:

	Filter 1: compactness (T^4 as a whole, from outside)	**Filter 2: fractality** (from inside, decompactified)
Effect	produces *discrete* spectrum	weights the modes unequally
forces	*existence* of sharp peaks	*location* of the dominant peaks
Origin	finite, closed topology	Jacobi degeneracy $g(\lvert n\rvert^2)$ + Bose-Einstein
Result	uniform mode lattice	peaks at $\lvert n\rvert^2 = 3, 6, 10, 14, \ldots$

The two are not two objects but two reference points on the same complete T^4: compact-finite from outside (Filter 1), fractally granulated inside down to $L_0 = \xi \cdot \ell_P$ (Filter 2). The fractal non-closure is therefore not a *second* property contradicting compactness, but the inside view of the same closed torus – a matter of reference point (inside/outside, R20).

Only both filters together give the observed structure:

- Filter 1 alone: uniform discrete lattice – every $|n|^2$ equal, no preferred peaks.
- Filter 2 alone: needs a discrete lattice to act on – no lattice without compactness.
- Filter 1 $+$ Filter 2: dominant peaks at specific $|n|^2$, which after 3D projection (Step 8) give the CMB series.

What can be deduced: The discrete peaks argue *for* a compact (finite) universe and *against* a purely-infinite-open one – a purely open space would give a continuum without sharp peaks. This is the strongest structural statement that can be drawn from the peaks.

Honest caveat: This conclusion is FFGFT-internal. The peaks do *not* force spatial compactness – ΛCDM produces the same discrete peaks from a *temporal* bound (the sound horizon at recombination) in a spatially infinite universe. The peaks alone do not distinguish spatial compactness (FFGFT) from a temporal bound (ΛCDM); this is the cosmological degeneracy (Doc. 267).

10.7 What the peak ratios state model-free

This section considers the measured peak positions purely with the methods of spectral / frequency analysis – without any T^4 assumption. The question: what can be read from the bare numbers $\ell \approx 220, 537, 810, 1120, 1450$, and where do typical misinterpretations from differencing and sampling limits lurk?

10.7.0 Clarification: "acoustic" is misleading

The CMB peaks are historically called "acoustic peaks". The term sends anyone who hears it for the first time in the wrong direction, because there is *nothing* to do with sound:

- **No sound:** there is no audible signal, no pressure wave in air.
- **No oscillation in time:** the CMB is a frozen still image, not a time signal. Nothing is measured in hertz.
- **No direct wave measurement:** the telescope measures the temperature as a function of *sky direction* – a map of hot and cold spots. What is evaluated is the *spatial distribution on the sky*, not an oscillation.

The word "acoustic" comes from the ΛCDM *explanatory model*: there the spots are interpreted as frozen pressure waves in the photon-baryon plasma *before* recombination. That is a model-dependent interpretation of the *mechanism* – it already sits in the name, before anything is measured. The *measured quantity* itself is neutral: a spatial interference pattern on the sky, described by the angular power spectrum C_ℓ.

Two different "frequencies" – do not confuse them. At the CMB two completely different things carry the name "frequency":

	(A) Photon frequency	(B) Spot pattern
What	colour/energy of the light	angular structure of the spots
Value	~ 160 GHz	$\ell = 220, 537, \ldots$
Unit	hertz (temporal)	dimensionless / angle (spatial)
measured as	radiometer (photon energy)	angle on the sky

The peak ratios $1 : 2.44 : 3.68 : \ldots$ belong to **(B)**, the *spatial* angular pattern. They have nothing to do with (A), the photon frequency. Converting the multipoles into a hertz frequency would be artificial and misleading – it would translate a spatial pattern into a fictitious temporal oscillation that does not exist.

Convention in this document: we avoid "acoustic" where possible and write *angular-spectrum peaks* or simply *CMB peaks*. The resonator analogy (open/closed pipe, double resonator) remains a purely *mathematical* analogy for a discrete eigenmode spectrum – not a claim that there was sound. Whether the resonances arise from pressure waves (ΛCDM) or from geometric lattice modes (FFGFT) is exactly the open question that the name "acoustic" prematurely decides in favour of ΛCDM.

10.7.1 First and second difference: the curvature is real

The spacings of successive peaks are *not* constant:

$$\Delta\ell = 317,\ 273,\ 310,\ 330.$$

A pure harmonic series (integer overtones $1 : 2 : 3 : 4 : 5$) would have constant spacings. The ratios $1 : 2.44 : 3.68 : 5.09 : 6.59$ grow *more slowly* than $1 : 2 : 3 : 4 : 5$ – the characteristic curve is **concave**. This curvature is much larger than the measurement uncertainty of the peak positions (~ 1–$3\,\%$ for broad peaks). It is therefore *not* a differencing artefact but a real feature of the spectrum.

Conclusion 1: integer overtones (a simple harmonic fundamental) are excluded.

10.7.2 The trap: "missing fundamental" when reading linearly

From the practice of frequency analysis it is known: if one measures only overtones and the fundamental is missing, the spacings suggest a fundamental that is not actually excited. Reading the CMB peaks linearly ($\ell_n = A + Bn$), the fit gives:

$$A = -72, \qquad B = 297.$$

The offset A is **negative**. The linear continuation thus demands a "fundamental" at $\ell < 0$ – physically meaningless. This is exactly the missing-fundamental artefact: whoever reads the peaks as "overtones of a shifted fundamental" constructs a fundamental mode that does not exist.

Conclusion 2: the linear reading (integer overtones plus offset) is a differencing artefact and is ruled out.

10.7.3 The fundamental in the lattice sense is reconstructible

If one instead checks whether the peaks are $\sqrt{\text{integer}}$ multiples of a common fundamental, i.e. $\ell = c \cdot \sqrt{|n|^2}$ with $|n|^2 = 3, 18, 42, 78$, then $c = \ell/\sqrt{|n|^2}$ is nearly constant:

$$c = 127.0;\ 126.6;\ 125.0;\ 126.8 \quad \text{(scatter } 1.6\,\%).$$

There thus exists a common **lattice fundamental** $c \approx 126$. This is the most parsimonious reading of the curvature: *one* parameter (the scale c) suffices, since the curvature itself sits parameter-free in the fixed lattice numbers $3, 18, 42, 78$.

	harmonic	**lattice ($\sqrt{\lvert n\rvert^2}$)**
Fundamental definition	$\ell_n = f_0 \cdot n$	$\ell = c \cdot \sqrt{\lvert n\rvert^2}$
Constancy test	$\ell/n = 220 \ldots 290$	$\ell/\sqrt{\lvert n\rvert^2} \approx 126$
Parameters	2 (offset + slope)	1 (scale c only)
Offset	$A = -72$ (unphysical)	none needed
Result	fails	$c \approx 126$

Conclusion 3: the fundamental is reconstructible – but only in the lattice sense ($\ell = c\sqrt{|n|^2}$), not the harmonic one. The value $c \approx 126$ corresponds to the mode $|n|^2 = 1$.

10.7.4 The reconstructed fundamental is not the first peak

The lattice fundamental $c \approx 126$ corresponds to $\ell \approx 126$ for $|n|^2 = 1$. The first *visible* peak, however, lies at $|n|^2 = 3$, i.e. $\ell \approx 219$. There is thus a **real but peak-invisible** fundamental mode:

$$|n|^2 = 1 \;\rightarrow\; \ell \approx 126 \quad \text{(fundamental, not a peak)},$$

$$|n|^2 = 3 \;\rightarrow\; \ell \approx 219 \quad \text{(first visible peak)}.$$

Remarkably, $\ell \approx 126$ lies *in the middle of the observed range*, but there is no peak there, rather the rising Sachs-Wolfe plateau. This is consistent with the observer mechanism (Step 8): the mode $|n|^2 = 1$ is the single-axis mode $(1, 0, 0, 0)$, which does *not* satisfy the symmetric 3D projection condition; only $(1, 1, 1, 0)$ with $|n|^2 = 3$ projects isotropically and becomes the first peak.

Conclusion 4: there is a real, lower-lying fundamental mode at $\ell \approx 126$ that acts as a scale-setter but, owing to missing symmetry, does not appear as a peak. This is a weakly falsifiable consistency condition: the fundamental c reconstructed from the upper peaks must match the place where the spectrum passes from the plateau into the first peak.

10.7.5 No size of the universe from the oscillations

No *absolute size* can be read from the angular oscillations – neither the size of the structures nor that of the universe. The measured angle of a peak is

$$\theta = \frac{L_{\text{structure}}}{D_A},$$

i.e. only the *ratio* of physical structure size $L_{\text{structure}}$ and distance D_A (angular distance to the source surface). The same angle fits "large structure, far away" just as well as "small structure, nearby":

assumed D_A	$\rightarrow L_{\text{structure}}$ **(at $\theta \approx 0.8^\circ$)**
$14\,000$ Mpc	≈ 200 Mpc
100 Mpc	≈ 1.4 Mpc
1 Mpc	≈ 0.01 Mpc

To turn the angle into an absolute length one needs D_A – and D_A is not a measured quantity but a **model output**:

- ΛCDM: $D_A = \int c/H(z)\,dz$ – requires H_0, Ω_m, Ω_Λ, the recombination z_*.
- FFGFT: $D_A = R_H \ln(1 + z_*)$ – requires R_H, the external parameter (R20), which does not follow from ξ alone.

In both cases the distance comes from the model, not from the measurement. The angular measurement is **scaleless**: it shows the *form* of the pattern (the ratios), but not its absolute size – like a photograph without a scale bar shows the form of an object, not its size. This is exactly the same limit as for the peak ratios: the *form* follows from the geometry, the *absolute scale* needs an external parameter (R20).

Why the resonator method does not help here. For a real resonator in front of us – an organ pipe, say – the size does sit in the fundamental: measuring the fundamental wavelength λ_0 *in metres* and knowing the medium (sound speed c_{medium}) gives the length directly, $L = \lambda_0/2$. This method needs two things, both of which are absent at the CMB:

- **A wavelength in metres** – not just an angle. The organ pipe is in front of us; we measure its length directly. The CMB "resonator" is unreachably far away; we see only its angle on the sky, not its length. The step from angle to length needs D_A – the unknown model output.
- **A known medium.** For the organ pipe c_{medium} is a known constant. In the ΛCDM plasma picture the "sound speed" $c_s(z) = c/\sqrt{3(1 + R(z))}$ is not a constant but a *function computed from the model*: it requires the baryon and photon densities as functions of time, the expansion history and the recombination state. Here the "medium" is not a substance one has in front of one and whose c one measures – it arises only *indirectly* from the assumed curvature and expansion. "Denser in the past" is already a statement about the geometry, not about an independently existing medium.

Thus in the ΛCDM picture the back-calculation is triply model-dependent: the "wavelength" is only an angle, the "medium" $c_s(z)$ is computed from density, expansion and metric, and the distance D_A is likewise a model output. None of the three ingredients is a model-free measurement; the "medium" is even the most derived quantity. FFGFT in turn needs *no* medium and *no* sound speed – the peaks are eigenmodes of a geometry, not sound in a substance – but does need R_H as an external parameter for the absolute scale (R20). Not assumption-free, but a different *kind* of assumption: a geometric scale instead of a whole plasma and expansion history.

10.7.6 What is certain and what is not – the sampling limit

Certainly readable (model-free):

- No integer overtones – the curvature is real, not an artefact.
- The curvature is concave (ratios grow sub-linearly).
- A discrete mode lattice with $\sqrt{|n|^2}$ scaling is the most parsimonious reading (one parameter, $c \approx 126$).
- There is no harmonic fundamental "below" the first peak; the linear fit there is a missing-fundamental artefact.
- No *absolute size* (of the universe or the structures) can be read – the angle gives only L/D_A, and D_A is a model output.

Not readable (limit of the method):

- Whether the concave curvature stems from a $\sqrt{\text{integer lattice}}$ (FFGFT) or from acoustic ΛCDM dispersion – both produce the same form (cosmological degeneracy, Doc. 267).
- Fine differences between curved laws: the CMB peaks are *broad* (half-width $\Delta\ell \sim 100$–200), the positions determinable only to ~ 1–$3\,\%$. This suffices to *exclude* integer overtones, not to distinguish the concave laws.

Conclusion of this section: pure frequency analysis – without any T^4 assumption – yields three robust statements: (i) no harmonic overtone series, (ii) the curvature is real and concave, (iii) a lattice fundamental $c \approx 126$ exists and corresponds to a real but invisible $|n|^2{=}1$ mode. These statements are obtained independently of the FFGFT derivation and support it from the measurement side, without removing the cosmological degeneracy (FFGFT vs. ΛCDM).

.13 Step 11: No fit – what is derived and what remains open

The agreement of the peak ratios to $< 2\%$ does *not* come from a fit. This section shows explicitly what comes from where.

11.1 What follows from ξ alone – and what does not

Correction relative to earlier versions

Earlier versions claimed the sequence $\{1, 6, 14, 26\}$ followed "from ξ alone as local maxima". This is *not* correct. The genuine local maxima of the full thermally weighted T^4 spectrum lie at $|n|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$ (dense). Of these, $6, 14, 26$ are local maxima, but 1 is *not*, and 10 (a local maximum) is missing from the sequence. So $\{1, 6, 14, 26\}$ is *not* a natural forward subset – it was read off the data (Step 4). What follows forward from T^4 is cleanly separated in Step 17.

Quantity	Origin
T^4 spectrum maxima at $\|n\|^2 = 3, 6, 10, 14, 18, 22, 26, 30, \ldots$	Local maxima of $I(r) = g(r) \cdot N(r) \cdot r$; g from the Jacobi formula (exact), N from Bose-Einstein. Input: only $\xi = 1/7500$. *Forward, but dense.*
Harmonic backbone $1 : 2 : 3 : 4 : 5$	Symmetric resonance (n, n, n): $k^2_{\text{obs}} = 3n^2$. *Forward*, but harmonic – "too clean".
Anharmonicity (stretch)	Codimension-1 projection $D_f - 1 \approx 2 \Rightarrow J_0$ Bessel. *Forward, parameter-free* (Step 17).
Selection of exactly $\{1, 6, 14, 26\}$	*Read off the CMB data* (Step 4), not forward.

Forward: the dense spectrum, the harmonic backbone and the geometric anharmonicity. **Backward:** the concrete selection of the four visible peaks.

11.2 What is not yet rigorously derived

Assumption	Status	Consequence
$L_{\text{tor}} = 1/(k_B T_{\text{CMB}})$	Physically motivated, not rigorously proved	Affects only the absolute scale, *not* the ratios
$k^2_{\text{obs}} = (n_1^2 + n_2^2 + n_3^2)/L^2$	Observer in 3D + time-unfolding, n_4 = time-unfolding	Affects the factor 3; physically motivated, mathematically open
D_A (angular distance)	External parameter R_H needed (R20)	Affects only $\ell_1 \approx 220$, not the ratios

11.3 What the ratios prove and what they do not

The ratios $1 : 2.449 : 3.742 : 5.099$ agree with the CMB to $< 2\%$ *without* free parameters. This shows:

- The T^4 geometry and the 3D observer projection *automatically* produce a peak structure of the right order of magnitude.
- The deviation of $< 2\%$ lies within the measurement uncertainty of the raw CMB peak positions (without a ΛCDM template fit).

It does *not* prove:

- That the selection mechanism (why the 3D observer prefers precisely the most symmetric modes with $n_1^2 + n_2^2 + n_3^2 = 3 \cdot |n_{\min}|^2$) follows rigorously from the spherical C_ℓ projection (still to be shown, R30).
- That no other theory can produce the same ratios.
- That FFGFT correctly describes the entire CMB (amplitudes, polarisation, early-phase physics open).

The status is: an unfitted result from ξ that matches the observed ratios to $< 2\%$. This is a genuine test – falsifiable by more precise model-free measurement of the raw peak positions.

.14 Step 12: Schwarzschild connection and R_H

12.1 The two Schwarzschild radii in FFGFT

FFGFT has two scales that are simultaneously Schwarzschild radii (Doc. 190, R4; Doc. 182):

Scale	Definition	Schwarzschild mass	Check
$L_0 = \xi \cdot \ell_P$	sub-Planck fundamental length	$M_0 = \xi \cdot m_P/2$	$r_S(M_0) = L_0$
$R_H = \hbar c/E_H$	Hubble horizon	$M_U = R_H c^2/(2G)$	$r_S(M_U) = R_H$

Both checks numerically:

$$r_S(M_0) = \frac{2G \cdot \xi \cdot m_P/2}{c^2} = \xi \cdot \frac{G m_P}{c^2} = \xi \cdot \ell_P = L_0 \quad ✓$$

$$r_S(M_U) = \frac{2G \cdot M_U}{c^2} = \frac{2G}{c^2} \cdot \frac{R_H c^2}{2G} = R_H \quad ✓$$

12.2 The conversion: R_H from the two Schwarzschild conditions

Dividing the two Schwarzschild formulas:

$$\frac{R_H}{L_0} = \frac{2G M_U/c^2}{2G M_0/c^2} = \frac{M_U}{M_0}$$

Thus:

$$\boxed{R_H = L_0 \cdot \frac{M_U}{M_0} = \xi \cdot \ell_P \cdot \frac{M_U}{\xi \cdot m_P/2} = \frac{2\,\ell_P \cdot M_U}{m_P}}$$

This is a pure conversion – no model, no assumption. It shows: R_H follows directly from L_0 and the mass ratio M_U/M_0.

12.3 What this means

$L_0 = \xi \cdot \ell_P$ is fully derived from ξ. $M_0 = \xi \cdot m_P/2$ likewise.

R_H is thus fully determined *once* M_U is known:

$$R_H = \frac{2\,\ell_P}{m_P} \cdot M_U$$

In Doc. 182 M_U is obtained via $E_H = E_0 \cdot \xi^{41/4}$. The Schwarzschild conversion shows the same R_H from another side – as a direct consequence of the mass scale of the universe.

12.4 Open question: M_U independently from ξ?

The connection $R_H = 2\ell_P M_U/m_P$ is exact. Open is: can M_U be derived from the T^4 mode structure *without* presupposing R_H?

If so, then R_H would be a genuine prediction from ξ alone – without needing the exponent $41/4$ as input. This is an open derivation, no fundamental obstacle.

.15 Step 13: The measurement problem – what FFGFT can do, what ΛCDM cannot

13.1 The exponent 41/4 is equivalent to H_0

From $E_H = E_0 \cdot \xi^{41/4}$ and $H_0 = E_H/\hbar$ it follows:

$$\frac{41}{4} = \frac{\ln(E_0/(\hbar H_0))}{\ln(1/\xi)}$$

The exponent $41/4$ is thus **equivalent to measuring** H_0. Numerically:

H_0 [km/s/Mpc]	Source	Exponent
66.19	FFGFT (from $41/4$)	10.2492
67.40	Planck/CMB (ΛCDM)	10.2472
73.00	SH0ES (distance ladder)	10.2382

The difference between the exponents is smaller than 0.02 – all H_0 measurements give the same exponent to $\sim 0.1\%$.

13.2 The measurement problem

There is no model-free measurement of H_0. Every measurement presupposes model assumptions:

Method	**Model assumption**	**Result**
CMB peaks (ΛCDM)	expansion, FLRW metric, recombination	67.4 km/s/Mpc
Distance ladder (SH0ES)	calibrated standard candles, local geometry	73.0 km/s/Mpc
FFGFT ($\xi^{41/4}$)	static universe, T^4 topology, exponent $41/4$ from theory	66.2 km/s/Mpc

In FFGFT H_0 means something different than in ΛCDM: not the expansion rate, but the reciprocal of the fractal path-lengthening scale $R_H = c/H_0$. What ΛCDM measures as expansion, FFGFT interprets as fractal path-lengthening $z(d)$ (Doc. 190, R14).

13.3 What FFGFT can do, what ΛCDM cannot

Quantity	ΛCDM	FFGFT
H_0	Free parameter (determined from data)	Derived from $\xi^{41/4}$ (if $41/4$ derived)
CMB peak ratios	From $r_s(z^*)$ and $D_A(z^*)$ (needs H_0, Ω_b, ...)	From T^4 geometry (no free parameter)
Absolute peak position ℓ_1	From H_0, z^*, r_s	From R_H (from $\xi^{41/4}$) + open connection R20

The structural difference: ΛCDM needs H_0 as input – measured, not derived. FFGFT derives $H_0 = 66.2$ km/s/Mpc from $\xi^{41/4}$. If $41/4$ follows from the T^4 geometry (still open), H_0 is a *prediction* – not a free parameter.

The remaining open problem: The exponent $41/4$ is not derived from the T^4 geometry in any document. It is the only place in FFGFT cosmology where an external input is needed – equivalent to H_0. As long as this derivation is missing, FFGFT and ΛCDM stand on equal footing in this respect: both need a cosmological scale as input.

.16 Step 14: The universal Compton-Schwarzschild relation

14.1 Every length has two associated masses

To every length R belong two masses:

$$M_c(R) = \frac{\hbar}{R\,c} \quad (R \text{ is its Compton wavelength})$$

$$M_s(R) = \frac{R\,c^2}{2G} \quad (R \text{ is its Schwarzschild radius})$$

Their product is *independent of* R:

$$\boxed{M_c(R) \cdot M_s(R) = \frac{\hbar}{Rc} \cdot \frac{Rc^2}{2G} = \frac{\hbar c}{2G} = \frac{m_P^2}{2}}$$

14.2 Application to the electron

Equivalently in lengths: to every mass m belong the Compton wavelength $\lambda = \hbar/(mc)$ and the Schwarzschild radius $r_S = 2Gm/c^2$, with

$$\lambda \cdot r_S = \frac{\hbar}{mc} \cdot \frac{2Gm}{c^2} = \frac{2G\hbar}{c^3} = 2\,\ell_P^2$$

Thus:

$$\sqrt{\lambda_e \cdot r_S(e)} = \sqrt{2}\,\ell_P$$

The two scales of the electron lie *geometrically symmetric* around the Planck length – independent of the electron mass.

Scale	Value	Meaning
$r_S(e) = 2Gm_e/c^2$	1.35×10^{-57} m	Schwarzschild (bottom)
ℓ_P	1.62×10^{-35} m	Planck (middle)
$\lambda_e = \hbar/(m_e c)$	3.86×10^{-13} m	Compton (top)

14.3 Application to R_H

The same relation holds for the Hubble radius:

$$M_c(R_H) = \frac{\hbar}{R_H c} = \frac{E_H}{c^2} \quad \text{(tiny)}$$

$$M_s(R_H) = \frac{R_H c^2}{2G} = M_U \quad \text{(mass of the universe)}$$

with

$$M_c(R_H) \cdot M_s(R_H) = \frac{E_H}{c^2} \cdot M_U = \frac{m_P^2}{2} \quad ✓$$

14.4 What this relation achieves and what it does not

It achieves: an elegant internal consistency. The Schwarzschild mass of the universe M_U and the cosmological energy E_H are connected by $M_U \cdot E_H/c^2 = m_P^2/2$ – exact, without free parameter. The two Schwarzschild scales L_0 (bottom) and R_H (top) are not arbitrary bounds but Schwarzschild radii of real masses ($M_0 = \xi m_P/2$ and M_U).

It does not achieve: a determination of R_H. The relation $M_c \cdot M_s = m_P^2/2$ is a universal identity – it holds for *every* length and therefore gives no new information about the numerical value of R_H. Both masses $M_c(R_H)$ and $M_s(R_H)$ are defined via R_H; their product is tautologically $m_P^2/2$.

To fix R_H one needs an *independent* input: the cosmological energy $E_H = E_0 \cdot \xi^{41/4}$ (Doc. 182). The electron fixes the Planck scale via $\sqrt{\lambda_e r_S(e)} = \sqrt{2}\ell_P$, but not R_H.

The remaining open problem stays: The exponent $41/4$ – equivalent to the cosmological energy scale E_H – is the only input that FFGFT needs for the absolute cosmological scale. The Compton-Schwarzschild relation explains the *structure* of the scale hierarchy, but not the *value* of E_H.

.17 Step 15: The T^4 length is a Schwarzschild radius

15.1 The fundamental identity (Doc. 010, 066, 067)

The FFGFT documents define the T0 length of an object as:

$$r_0 = 2Gm \quad \text{(natural units, } c = 1\text{)}$$

Doc. 067 explicitly states: the factor 2 results from the relativistic field structure and *coincides exactly with the Schwarzschild radius*. Doc. 010 writes $r_0 = 2GE$ (energy instead of mass), Doc. 066 as the ratio $\xi = r_0/\ell_P = 2Gm/\ell_P$.

This means: every characteristic T0 length *is* a Schwarzschild radius – not analogous, but identical.

15.2 Why different formulas in the documents?

It is the same formula in different unit systems:

Document	Formula	Unit system
SI	$r_S = 2Gm/c^2$	metres, kg, seconds
Doc. 015, 029	$r_S = 2M$	natural ($G = c = 1$)
Doc. 010	$r_0 = 2GE$	$c = 1$, energy instead of mass
Doc. 066	$r_0 = \xi \cdot \ell_P$	Planck ($\ell_P = 1$)

For the minimal mass $M_0 = (\xi/2)\, m_P$ *all* give the same value:

$$\frac{2GM_0}{c^2} = \frac{2G}{c^2} \cdot \frac{\xi}{2}\, m_P = \xi \cdot \frac{G m_P}{c^2} = \xi \cdot \ell_P = L_0 = 2.155 \times 10^{-39}\ \text{m}$$

The apparent variety is only the SI conversion, which shows or hides factors c^2 and G. As soon as one computes in SI units, c^2 and G appear explicitly; in natural units they vanish. The physics is identical in all cases.

15.3 "a Schwarzschild radius" vs. "the Schwarzschild radius"

In standard physics one speaks of *the* Schwarzschild radius of an object – the one length at which this mass becomes a black hole.

In FFGFT the T0 length is *always* a Schwarzschild radius – but *which one* depends on the mass:

Mass	Schwarzschild radius	Role
Electron m_e	1.35×10^{-57} m	particle scale
$M_0 = (\xi/2)m_P$	$L_0 = 2.16 \times 10^{-39}$ m	fundamental length
Proton	2.48×10^{-54} m	particle scale
M_U (universe)	$R_H = 1.41 \times 10^{26}$ m	Hubble horizon

Therefore: L_0 *is a* Schwarzschild radius (namely that of M_0) – but *not the* Schwarzschild radius as such, because there is no single universal one. Likewise R_H is *a* Schwarzschild radius (that of M_U), not *the* one.

This is the scale relativity from Doc. 182: every system carries its own torus with its own characteristic scale $R_{\text{torus}}(m) = \hbar/(mc)$. The universal lower bound L_0 holds for all; a universal upper bound does not exist – R_H is only the upper bound of the cosmological sector.

15.4 What this means for R_H

R_H *coincides exactly* with the Schwarzschild radius of M_U – this is correctly stated in the documents and is not an approximation. But: because $r_0 = 2Gm$ is the *definition* of the T0 length, the agreement is necessary, not surprising. The T0 length of a mass and its Schwarzschild radius are the same object.

The identity does not fix R_H independently: one still needs M_U (resp. $E_H = E_0 \cdot \xi^{41/4}$). What the identity achieves is the *physical interpretation*: R_H is not an abstract horizon but the Schwarzschild radius of the total mass of the static universe – the universe sits at its own Schwarzschild bound.

.18 Step 16: Why the Schwarzschild relation does not derive R_H

16.1 The tempting but false conclusion

From two true formulas a fallacy arose:

$$\text{(A)} \quad \xi = L_0/\ell_P, \quad L_0 = \xi \cdot \ell_P \tag{1}$$

$$\text{(B)} \quad r_S(m) \cdot r_C(m) = 2\,\ell_P^2 \quad \text{(universal)} \tag{2}$$

Both are correct. The error arose through a tacit assumption: that in relation (B) one may replace the electron mass m_e by the universe mass M_U *while keeping* λ_e, thus obtaining $R_H = r_S(M_U)$.

16.2 The exact error

Relation (B) connects two quantities of the *same* mass. For the electron:

$$\underbrace{r_S(m_e)}_{1.35\times10^{-57}\,\text{m}} \cdot \underbrace{\lambda_e}_{3.86\times10^{-13}\,\text{m}} = 2\,\ell_P^2$$

Both factors belong to m_e.

Replacing $m_e \to M_U$ replaces *both* factors:

$$\underbrace{r_S(M_U)}_{R_H = 1.41\times10^{26}\,\text{m}} \cdot \underbrace{r_C(M_U)}_{3.7\times10^{-96}\,\text{m}} = 2\,\ell_P^2$$

The Compton wavelength of M_U is 3.7×10^{-96} m – **not** λ_e. The conflation was: R_H (Schwarzschild of the *universe*) was put together with λ_e (Compton of the *electron*) into a relation that holds only for a single mass. In fact:

$$r_S(M_U)\cdot\lambda_e = 5.4\times10^{13}\,\text{m}^2 \neq 2\,\ell_P^2 = 5.2\times10^{-70}\,\text{m}^2$$

– off by a factor of 10^{83}.

16.3 Two separate systems

Electron and universe are two different systems, each consistent in itself:

$$\text{Electron:}\quad m_e \to \{r_S(e),\,\lambda_e\},\quad r_S(e)\cdot\lambda_e = 2\ell_P^2$$

$$\text{Universe:}\quad M_U \to \{R_H,\,r_C(M_U)\},\quad R_H\cdot r_C(M_U) = 2\ell_P^2$$

Between the systems there is *no* bridge from this relation.

16.4 The Schwarzschild formula generates no information

The deeper reason: $r_S = 2GM/c^2$ is a *conversion* between mass and length. As for any object:

Known	Follows
Mass M	Schwarzschild radius $r_S = 2GM/c^2$
Length r_S	Mass $M = r_S c^2/(2G)$
neither	**nothing**

This holds for a black hole, the Sun, the electron and the universe alike. To know R_H, one of the four equivalent quantities must be independently known:

$$M_U \longleftrightarrow H_0 \longleftrightarrow E_H \longleftrightarrow R_H$$

They form a ring of conversions – but no entry point.

16.5 Where FFGFT supplies the input

FFGFT supplies exactly one cosmological input:

$$E_H = E_0\cdot\xi^{41/4},\qquad E_0 = \sqrt{m_e\,m_\mu}$$

From this follows $R_H = \hbar c/E_H$ and everything else. The exponent $41/4$ is equivalent to specifying H_0 and is not derived from the T^4 geometry in any document (open problem).

Conclusion: The Schwarzschild identity (L_0, R_H are Schwarzschild radii) is a genuine structural property and supplies the *physical interpretation*. But it cannot *generate* the cosmological scale – for that the one input E_H is needed. The earlier impression that the two formulas (A) and (B) would derive R_H rested on the tacit substitution $m_e \to M_U$, which conflates two different masses.

.19 Step 17: Forward test – anharmonicity from the codimension-1 projection

This step cleanly separates what follows *forward* (without looking at the data) from T^4 and what is a *backward* consistency check. It documents an explicit forward investigation and is the conceptual advance over Steps 1–11. All numbers here are reproducible with the accompanying scripts (`Dok267_268_Skripte/`).

17.1 What the naive forward computation yields – and what it does not

Three forward computations, all without any look at the data:

1. **Symmetric T^4, spherical C_ℓ Bessel projection** ($C_\ell = \sum_n N_{\text{BE}} \, |j_\ell(k_{\text{obs}} D_A)|^2$): does *not* reproduce $\{3, 18, 42, 78\}$. The result is a quasi-continuum dominated by the lowest modes (strongest peak at $\ell \approx 122$). The naive Bessel projection is *insufficient* (contrary to the earlier P30 wording "a feasible computation, no fundamental obstacle").
2. **3+time (T^3 spatial spectrum):** local maxima at $|n|^2 = 2, 5, 9, 14, 17, 21, 26, \dots$ – dense, and *not* $\{1, 6, 14, 26\}$.
3. **Symmetric spatial resonance** (n, n, n)**:** $k^2_{\text{obs}} = 3n^2 \Rightarrow$ series $\{3, 12, 27, 48, 75\}$, ratios $\sqrt{3} : \sqrt{12} : \sqrt{27} : \dots = 1 : 2 : 3 : 4 : 5$ (*harmonic*).

Conclusion: forward, T^4 produces either a dense spectrum or the harmonic backbone $1 : 2 : 3 : 4 : 5$. The sparse, anharmonic series $\{3, 18, 42, 78\}$ is *not* a forward output; it arises only from the data-read selection $\{1, 6, 14, 26\}$.

Concretely open ($|n|^2 = 30$**, sharpened by O. Teker):** $30 = 3 \cdot 10$, and $|n|^2 = 10$ is a genuine local maximum ($I = 20.1$). So 30 satisfies the factor-3 filter. Thermally it is even *stronger* than two of the visible peaks: $I(30) = 13.2 > I(42) = 7.6 > I(78) = 1.7$. Yet the CMB shows no peak at $|n|^2 = 30$ ($\ell \approx 696$, in the trough between peaks 2 and 3). The "strongest in range" argument (earlier version) fails here. The selection mechanism is thus *not* supplied by the factor 3 alone (P29 only partial, P30 open).

17.2 The actual forward finding: anharmonicity is geometric

The harmonic backbone $1 : 2 : 3 : 4 : 5$ is "too clean" – like the ideal overtones of a stiffness-free string. Real strings are *inharmonic* due to their stiffness (stretched piano tuning). By analogy we *expect* anharmonicity, because spacetime is curved from our viewpoint and the projection is nonlinear (fractal path lengthening).

Acoustic resonators show that anharmonicity is purely *geometric* (forward, no fit, no medium):

Resonator (dimension D)	**Ratios**	**Dev. to CMB**
Tube/sphere j_0 ($D = 3$, $\nu = \frac{1}{2}$)	$1 : 2 : 3 : 4$ (harmonic)	-18 to -24%
Circular membrane J_0 ($D = 2$, $\nu = 0$)	$\mathbf{1 : 2.30 : 3.60 : 4.90}$	$\mathbf{-2}$ to $\mathbf{-6}\%$
S^3 Laplacian	$1 : 1.63 : 2.24$	-33 to -48%
S^2 ($Y_{\ell m}$)	$1 : 1.73 : 2.45$	too dense
CMB observed	$1 : 2.44 : 3.68 : 5.09$	—

Key fact: the radial Bessel index is $\nu = D/2 - 1$. Anharmonicity is *maximal at* $D = 2$ ($\nu = 0$, J_0) and vanishes at $D = 1$ and $D = 3$ (both harmonic). The CMB series sits exactly at $D \approx 2$ – the dimension of maximal anharmonicity.

Important: the match is to the *flat* circular membrane (J_0), *not* to the curved 2-sphere ($Y_{\ell m}$, which gives $1 : 1.73 : ...$, wrong). The correct expectation is a *flat bounded disk*, not a sphere – because the CMB peaks come from the *radial* projection, not from the eigenvalues of the observation surface.

17.3 Derivation: flat disk from the fractal codimension-1

The last-scattering surface is a **codimension-1 hypersurface** in the fractal space:

$$D_{\text{eff}} = D_f - 1 = (3 - \xi) - 1 = 2 - \xi \approx 2.$$

Radial modes in D dimensions are zeros of $J_{D/2-1}$. For $D = 2 - \xi$:

$$\nu = \frac{D}{2} - 1 = \frac{2 - \xi}{2} - 1 = -\frac{\xi}{2} \approx -6.7 \times 10^{-5} \approx 0 \;\Rightarrow\; J_0 \text{ membrane.}$$

This gives, forward, **parameter-free and without mode selection**, the anharmonic series $1 : 2.30 : 3.60 : 4.90$ – a match to $\sim 5\%$. The anharmonicity follows from the codimension-1 projection in the fractal geometry, exactly as expected from the curvature.

17.4 The small residual – honestly open

The best fit lies at $\nu \approx -0.07$, i.e. effective dimension $D \approx 1.86$ – just *below* 2. The obvious physical source would be the *spectral dimension* d_s (governing wave propagation on a fractal), which is generally below the Hausdorff dimension: $d_s = 2D_f/d_w$ with walk dimension $d_w > 2$.

The explicit computation rules this out, however:

- The FFGFT fractal is *too weakly fractal*: with $D_f = 3 - \xi$, $d_w = 2 + O(\xi)$, so $d_s(\text{surface}) \approx 2.00$. ξ is $\sim 1000\times$ too small to push D down to 1.86.
- For $d_s = 1.86$ one would need $d_w \approx 2.15$ (strong fractality), incompatible with $D_f = 2 - \xi$.

The residual is not derivable from ξ. Moreover it is of the order of the measurement uncertainty: the pure J_0 prediction ($D = 2$) is consistent with peaks 3 and 4 within $\sim 2\sigma$; only peak 2 sits at $\sim 4\sigma$. The "best fit 1.86" is thus largely measurement noise; pointing to an exact value (e.g. $13/7$) would be numerology unsupported by the data quality.

17.5 Summary – layered, honest structure

Layer	Status
$D = 3 \to 2$ (codimension-1, J_0)	**forward, parameter-free, derived**; $\sim 5\%$
Anharmonicity geometric	derived (maximal at $D = 2$)
Ratio match $\{3, 18, 42, 78\}$	retrodiction (from data), *not* prediction
$D = 2 \to 1.86$ (sub-2)	not from ξ; largely measurement noise; open
Exclusion of $\lvert n\rvert^2 = 30$	open (P29 partial, P30)
Peak-2 residual ($\sim 4\sigma$)	small genuine open point, likely other physics

Defensible statement (instead of "fit to $D = 1.86$"): the fractal codimension-1 projection produces, forward and parameter-free, the J_0 membrane spectrum ($D \approx 2$) and matches the CMB anharmonicity to $\sim 5\%$ – within the measurement errors. The harmonic backbone comes from the symmetric T^4 resonance; the stretch from the geometry of observation. This formulation is conceptually cleaner than the earlier mode selection because it requires *no* selection from the data, thereby removing the circularity objection (O. Teker). The rigorous selection mechanism for the discrete individual peaks remains open.